

Module 4: Noise in Comm Systems



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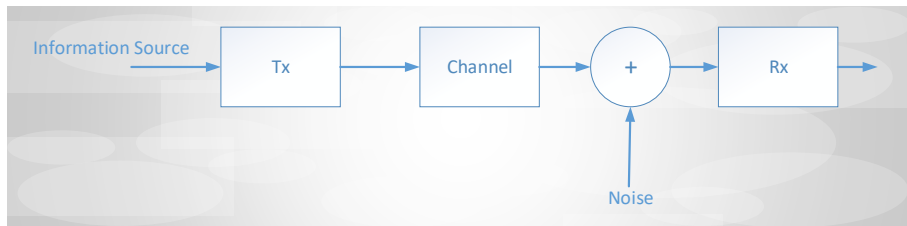
Objectives: M4L25

- Random Variables
- Q function

Randomness in Communications

- Two types of randomness
- Desired Signal (random but deterministic in a sense)
- Noise (Random all the way)
- Interference: a type of noise (can be deterministic if the sources are known)

Model of Communication System Under Noise



$$y = h * x + w$$

Information source and noise are random

How to characterize them mathematically in a tractable form?

Random Variables

- A random variable X is a mapping from the sample space to the set of real numbers.
- Notation: Capital letters (e.g., X) for random variables and small letters (e.g., x) to the value the random variable it takes.
- The probability of a RV X taking a value x_i is $P_X(x_i) =$ Probability of “ $X = x_i$.”

Cumulative Distribution Function (CDF)

- The cumulative distribution function (CDF) $F_X(x)$ of a RV X is the probability that X takes a value less than or equal to x

$$F_X(x) = P(X \leq x)$$

- Properties of CDF:
- $F_X(x) \geq 0$
- $\lim_{x \rightarrow \infty} F_X(x) = 1$ and $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- $P(a < X \leq b) = F_X(b) - F_X(a)$.

Probability Distribution Function (PDF)

- Definition

$$f_X(x) = \frac{d}{dx}F_X(x)$$

- Properties of PDF:

- $f_X(x) \geq 0$
- $\int_{-\infty}^{\infty} f_X(x)dx = 1$
- $P(a < X \leq b) = \int_a^b f_X(x)dx.$
- $F_X(x) = \int_{-\infty}^x f_X(x)dx$

Uniform Random Variable

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

Find the CDF?

Find the mean?

Find the variance?

Gaussian or Normal Random Variable

Denoted by $N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Which signal in communications can represent Gaussian random variable?

Gaussian Random Variable: Example

- Find the CDF of the standard Gaussian random variable
- $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}$
- $F_X(x) = \int_{-\infty}^x f_X(x) dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{\frac{-t^2}{2}} dt$
- $F_X(x) = P(X \leq x) = 1 - P(X > x)$
- Q-function: $Q(x) = P(X > x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{\frac{-t^2}{2}} dt$
- Q-function computes the tail probability of the Gaussian random variable.
- $Q(0) = 0.5$; $Q(\infty) = 0$
- Chernoff Bound: $Q(x) \leq \frac{1}{2} e^{\frac{-x^2}{2}}$
- For general Gaussian: $P(X > x) = Q(\frac{x-m}{\sigma})$

Q-Function

- Let $n(t) \sim N(\mu, \sigma^2)$. Find $P(n(t) > A)$
- What is $Q(0)$?
- What is $Q(\infty)$?
- What is a bound on $Q(x)$

- $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{t^2}{2}} dt$
- $erfc(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt = 2Q(x\sqrt{2})$
- $Q(x) = \frac{1}{2}erfc(\frac{x}{\sqrt{2}}) = \frac{1}{2}[1 - erf(\frac{x}{\sqrt{2}})]$
- $erf(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$

Statistical Average

- Expected value: $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$
- $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$
- Can you see the difference/similarity?
- Variance $\sigma_X^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
- $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, if X and Y are independent.

Sum of Independent Random Variables

- Let $X_1, X_2, \dots, X_n$ are independent RV.
- $Z = X_1 + X_2 + X_3 + \dots + X_n$
- $f_Z(z) = f_{X_1}(x) * f_{X_2}(x) * f_{X_3}(x) * \dots * f_{X_n}(x)$

Sum of Gaussian Random Variables

- Let $X_1, X_2, \dots, X_n$ are independent Gaussian RV.
- $Z = X_1 + X_2 + X_3 + \dots + X_n$: is still Gaussian
- Mean: $\mu(z) = \mu(X_1) + \mu(X_2) + \dots + \mu(X_n)$
- Variance: $\sigma^2(z) = \sigma^2(X_1) + \sigma^2(X_2) + \dots + \sigma^2(X_n)$

Central Limit Theorem

- Let $X_i, i = 1, 2, \dots, N$ be set of random variables which are statistically independent.
- The X_i have the same probability distribution with mean μ_x and variance σ_X^2 .
- Define a normalize random variable: $Y_i = \frac{1}{\sigma_X}(X_i - \mu_X), i, 1, 2, \dots, N$.
- $\mathbb{E}[Y_i] = 0$ and $\text{var}[Y_i] = 1$.
- Define a random variable as: $V_N = \frac{1}{\sqrt{N}} \sum_{i=1}^N Y_i$
- Central limit theorem:
- Probability distribution of V_N approaches a normalized Gaussian random variable $N(0, 1)$ as N approaches infinity.

Objectives: M4L26

- Random Process

Problems

Find $P(X > x)$ if $X \sim N(0, 1)$

Find $P(X > x)$ if $X \sim N(\mu, \sigma^2)$

Find $P(X > x)$ if $X \sim U(-1, 1)$

Show that $Q(-x) = 1 - Q(x)$

Find $P(-X > x)$ if $X \sim N(0, 1)$

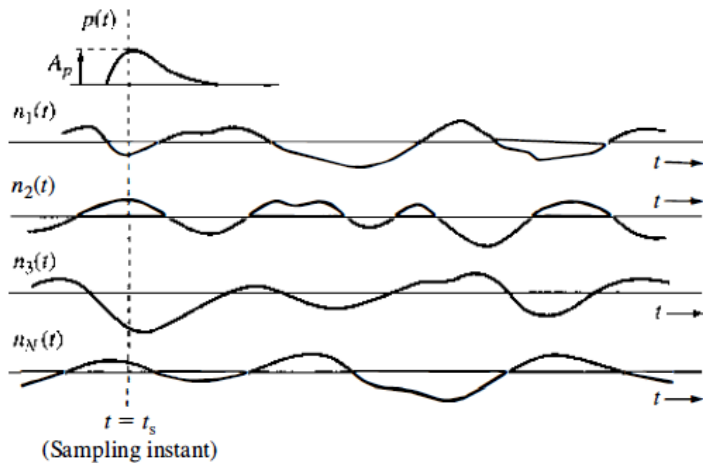
The noise voltage in an electronic circuit can be modeled as a Gaussian random variable $N(0, 10^{-8})$. Find the probability that the noise voltage is between -2×10^{-4} and 10^{-4} .

Answer: $1 - Q(1) - Q(2)$

Random Process or Stochastic Process

- Random process or Stochastic Process: A random variable which is a function of time.
- Collection of random variables at $t_1, t_2, \dots$ $\{X(t_1), X(t_2), \dots\}$.
- Examples: Signal transmissions as well as noise: both are random and vary with time

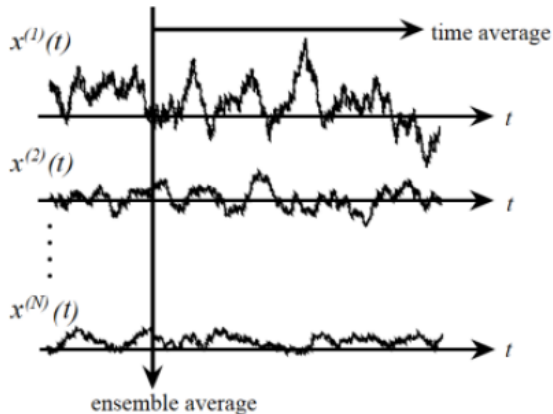
Noise Process



Random Process: Sketch

- TB 6.8-1 $x(t) = a \cos(2\pi f_c t + \theta)$, a is uniform RV $(-A, A)$
- TB 6.8-3 $x(t) = at + b$, a is uniform RV $(-2, 2)$, b is constant
- TB 6.8-7 $x(t) = kt$, where k is an RV in the interval $(-1, 1)$

Ensemble Average versus Time Average



Expectation of Random Process

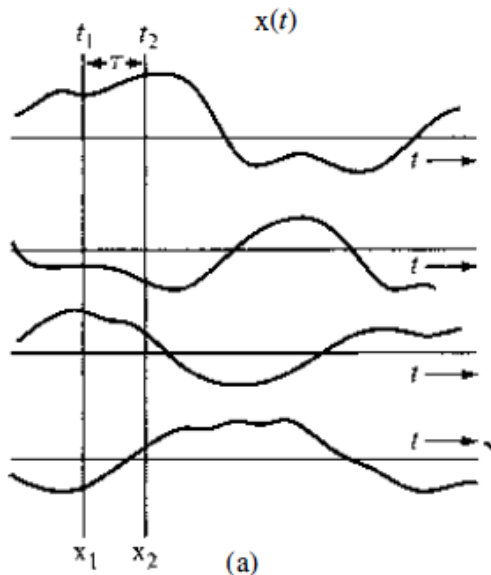
- Deterministic function of time.
- $m_X(t) = E[X(t)]$, for all t .
- $E[X(t_0)] = m_X(t_0) = \int_{-\infty}^{\infty} x f_{X(t_0)}(x) dx$

Expectation of Random Process: Example

- Find mean of $x(t) = A \cos(2\pi f_0 t + \phi)$, where $f_\phi = 1/2\pi$ in the interval of 0 and 2π
-
- TB 6.8-3 $x(t) = at + b$, a is uniform RV $(-2, 2)$, b is constant. Find mean of $x(t)$.
- TB 6.8-7 $x(t) = kt$, where k is an RV in the interval $(-1, 1)$. Find mean of $x(t)$.

Autocorrelation Function: Random Process

- Used to describe the PSD of random processes.
- $R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = \int_{-\infty}^{\infty} x_1 x_2 f_{X(t_1), X(t_2)}(x_1, x_2) dx_1 dx_2$
- ACF is a deterministic function of two variables t_1 and t_2 .



Objectives: M4L27

- AWGN

Example: Autocorrelation Function

- $X(t) = A \cos(2\pi ft + \phi)$
where ϕ is uniform RV between 0 to 2π

Example: Autocorrelation Function

- $X(t) = A \cos(2\pi ft + \Phi)$
- $R_{XX}(t_1, t_2) = A^2 E[\cos(2\pi ft_1 + \Phi) \cos(2\pi ft_2 + \Phi)]$
- $R_{XX}(t_1, t_2) = \frac{A^2}{2} \cos(2\pi f(t_1 - t_2))$

Example: Autocorrelation Function

- TB 6.8-3 $x(t) = at + b$, a is uniform RV $(-2, 2)$, b is constant. Find ACF of $x(t)$.
- TB 6.8-7 $x(t) = kt$, where k is an RV in the interval $(-1, 1)$. Find ACF of $x(t)$.

Example

If $x(t)$ and $y(t)$ are two WSS process, find $R_u(\tau)$, $R_v(\tau)$, $R_{uv}(\tau)$, and $R_{vu}(\tau)$, where $u(t) = 2x(t) - y(t)$ and $v(t) = x(t) + 3y(t)$.

Solution

- $R_{uv}(\tau) = E\{u(t)v(t + \tau)\}$
- $R_{uv}(\tau) = E\{[2x(t) - y(t)][x(t + \tau) + 3y(t + \tau)]\}$
- $R_{uv}(\tau) = E\{2x(t)x(t + \tau)\} - 3E\{y(t)y(t + \tau)\}$
- $R_{uv}(\tau) = 2R_x(\tau) - 3R_y(\tau)$

Wiener–Khinchin theorem: PSD of Random Process

- Fourier transform of ACF
- For stationary process: $S_x(f)$ and $R_x(\tau)$ are FT pairs.

Types of Process: Wide-sense Stationary Process

- $m_X(t) = E[X(t)]$ is independent of t .
- $R_{XX}(t_1, t_2)$ depends only on the time difference $\tau = t_1 - t_2$ and not on t_1 and t_2 individually.

Types of Process: Wide-sense Stationary Process

-
- $X(t) = A \cos(2\pi f_t + \Phi)$
- $R_{XX}(t_1, t_2) = \frac{A^2}{2} \cos(2\pi f(t_1 - t_2))$
- $m_x(t) = 0$
- $X(t)$ is WSS.

Random Process through Linear System

- If a stationary process $X(t)$ is passed through a linear system $h(t)$, show the following:
- $m_Y = m_X \int_{-\infty}^{\infty} h(t) dt$
- $R_y(\tau) = R_x(\tau) * h(\tau) * h(-\tau)$
- $S_y(f) = S_x(f) |H(f)|^2$

Additive White Gaussian Noise

- Unwanted signals: beyond our control (till date)
- Sources of noise: external (atmospheric noise, man-made noise, etc.)
- Sources of noise: internal (spontaneous fluctuations of current or voltage in electrical circuits).
- Shot noise and Thermal noise

Shot Noise

- Shot noise: arises in electronic circuits such as diodes and transistors because of the discrete nature of charge carriers.
- Also known as Schottky noise
- Fluctuations of discrete charge carriers around a mean value in successive-time interval is called shot noise.
- carriers: discrete.
- Number of charge carriers vary randomly.

Thermal Noise

- Random motion of electrons in passive resistors.
- Also called Johnson noise.
- Fluctuations caused by electron-atom collisions.
- Mean kinetic energy $\frac{1}{2}kT$, $k = 1.38 \times 10^{-23}$ J/K: Boltzmann's constant and T is absolute temperature.
- Maximum noise power in the frequency band B : $N_0 = kTB$
- Half-side power = $N_0/2 = \frac{kT}{2}$
- Two sided Noise PSD: $S_n(f) = \frac{kT}{2}$:
- Autocorrelation: $R_n(\tau) = \frac{N_0}{2}\delta(\tau)$

Thermal Noise: Example

- Calculate noise power at room temperature for a $BW = 1\text{Hz}$.
- $N_0 = kTB$

Noise PSD: Both-Sided

- Both sided versus single sided

Objectives: M4L28

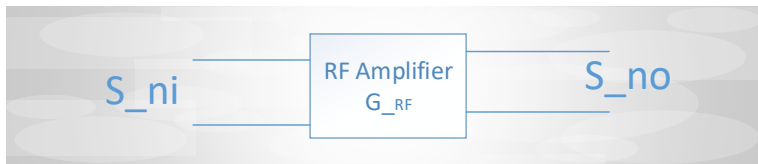
- Noise Figure

Noise Temperature and Noise Figure

- Noise temperature of a device: $T = \frac{P_n}{kB}$
- Noise figure of a system:
$$F = \frac{\text{Total noise power at the output}}{\text{Output noise power solely due to the input noise power}}$$
- Noise figure of a system:
$$F = \frac{\text{Total noise PSD at the output}}{\text{Output noise PSD assuming the system noiseless}}$$

Noise Figure

- RF amplifier: internal noise PSD $S_{n,RF}(f)$
- Input noise PSD: $S_{ni}(f)$



- Output noise PSD: $S_{no}(f) = S_{ni}(f)G_{RF}(f) + S_{n,RF}(f)$
- Noise figure

$$F(f) = \frac{S_{no}(f)}{S_{ni}(f)G_{RF}(f)} = \frac{S_{ni}(f)G_{RF}(f) + S_{n,RF}(f)}{S_{ni}(f)G_{RF}(f)} = 1 + \frac{S_{n,RF}(f)}{S_{ni}(f)G_{RF}(f)}$$

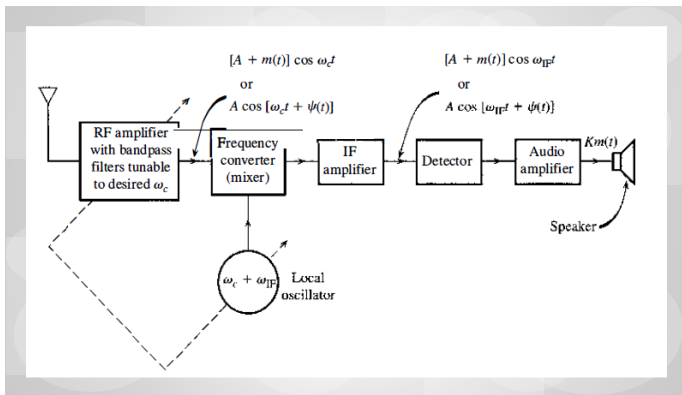
Equivalent Noise Temperature: Single Device

Prove that

- $T_{RF} = (F_{RF} - 1)T_0$
- $T_0 = 290$ degree Kelvin.

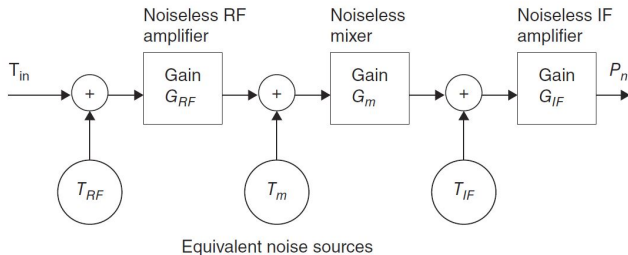
Noise Characterization of Receiver System: Multiple Devices

- Superhetrodyne receiver



Noise Characterization of Receiver System

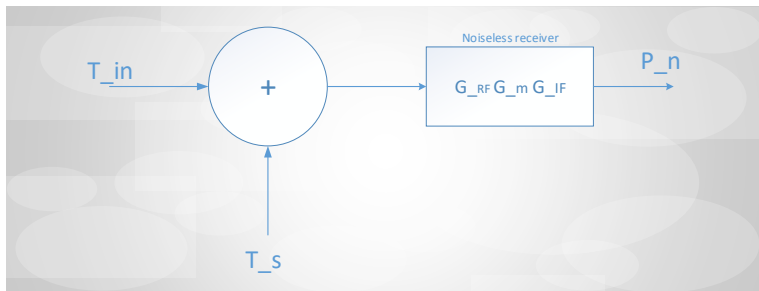
- Equivalent noise temperatures with noiseless systems:



Noise Characterization of Receiver System

- Superhetrodyne receiver
- $P_n = G_{IF}kT_{IF}B + G_{IF}G_m kT_m B + G_{IF}G_m G_{RF}k(T_{RF} + T_{in})B$
- Assume $T_{in} = 0$
- $P_n = kB G_{IF}G_m G_{RF}[(T_{RF}) + \frac{T_m}{G_{RF}} + \frac{T_{IF}}{G_{IF}G_{RF}}]$
- $T_s = [(T_{RF}) + \frac{T_m}{G_{RF}} + \frac{T_{IF}}{G_{RF}G_m}]$

Noise Characterization of Receiver System: Equivalent System



Noise Characterization of Receiver System: Noise Figure of Equivalent System

- $T_s = T_{RF} + \frac{T_m}{G_{RF}} + \frac{T_{IF}}{G_{RF}G_m}$
- $T_s = (F - 1)T_0$
- $T_{RF} = (F_{RF} - 1)T_0$ and so on.
- Substituting
- $F = F_{RF} + \frac{F_m - 1}{G_{RF}} + \frac{F_{IF} - 1}{G_{RF}G_m}$

Noise Figure of Equivalent System: General System

- $T_s = [T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} + \dots + \frac{T_n}{G_1 G_2 \dots G_{n-1}}]$
- $F = [F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots + \frac{F_n - 1}{G_1 G_2 \dots G_{n-1}}]$

TB Problem: 7.1-6

7.1-6 Three amplifiers A , B , C have following specifications

$$F_A = 2$$

$$G_A = 100$$

$$F_B = 2$$

$$G_B = 10$$

$$F_C = 4$$

$$G_C = 100$$

Obtain the noise figure values for the combination (a) $A-B-C$ and (b) $B-A-C$. Comment on the results obtained. Note that A and B have the same noise figure value.

Solution

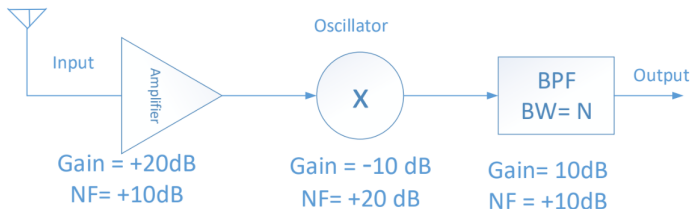
$$F_{ABC} = 2 + \frac{2-1}{100} + \frac{4-1}{1000} = 2.013$$

$$F_{BAC} = 2 + \frac{2-1}{10} + \frac{4-1}{1000} = 2.103$$

ABC is better!

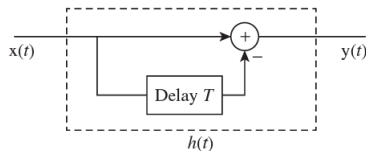
CW Problem

A signal has an input power spectral density 1W/Hz . Find the power of the signal at the output of the BPF.



CW Problem

A WSS random process $x(t)$ is passed through the system as shown in figure. Obtain the relationship between $S_x(f)$ and $S_y(f)$.

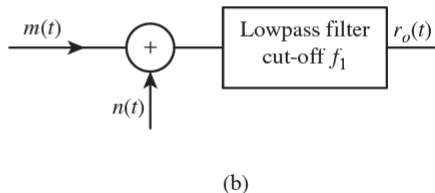
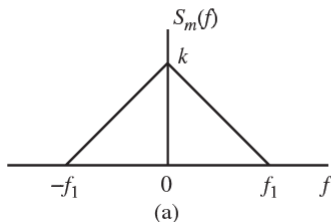


Solution

- $y(t) = x(t) - x(t - T)$
- $h(t) = \delta(t) - \delta(t - T)$
- $H(f) = 1 - e^{-j2\pi fT}$
- $H(f) = 1 - (\cos 2\pi fT - j \sin 2\pi fT) = (1 - \cos 2\pi fT) + j \sin 2\pi fT$
- $S_y(f) = S_x(f)|H(f)|^2$
- $S_y(f) = |(1 - \cos 2\pi fT) + j \sin(2\pi fT)|^2 S_x(f) = [2 - 2 \cos 2\pi fT] S_x(f)$

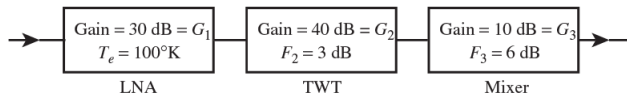
Figure of Merit: Analog Modulation

The PSD of signal $m(t)$ is given in the figure. Find the SNR at the output if the noise PSD is $N/2$ (two-sided).



CW Problem

Obtain the effective noise temperature for the ground receiver shown in Fig. P7.1-7. Hence, determine the noise figure of the receiver.



Solution

- $G_1 = 30\text{dB} = 10^3$.
- $G_2 = 40\text{dB} = 10^4$.
- $G_3 = 10\text{dB} = 10$.
- $T_{e1} = 100\text{K}$, $F_2 = 3\text{dB} = 2$; $F_3 = 6\text{dB} = 4$
- $T_{e2} = (F_2 - 1) \times 290 = 290\text{K}$
- $T_{e3} = (F_3 - 1) \times 290 = 870\text{K}$
- $T_e = T_{e1} + \frac{T_{e2}}{G_1} + \frac{T_{e3}}{G_1 G_2}$
- $T_e = 100 + \frac{290}{10^3} + \frac{870}{10^7} = 100.377$
- Noise Figure: $F = 1 + \frac{T_e}{T_0} = 1.346$
- Or, $F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} = 1.346$

Figure of Merit

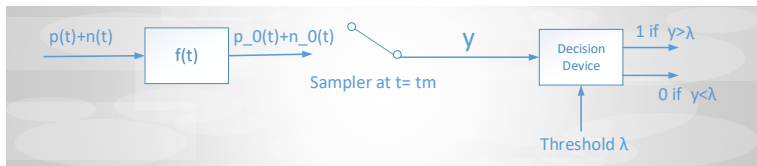
$$FoM \triangleq \frac{\text{SNR}_{\text{out}}}{\text{SNR}_{\text{in}}}$$

where SNR_{in} is the pre-detection SNR at the receiver input (and SNR_{out} is the post-detection SNR at the demodulator output).

FoM for DSB-SC

Digital Receiver Design and BER Performance

Receiver: $f(t)$



- Determine the filter $f(t)$ that minimizes the BER.
- $y(t_m) = \pm p_0(t_m) + n_0(t_m)$.
- What is BER?
- $P_e = Q\left(\frac{A_p}{\sigma_n}\right)$
- $P_e = Q\left(\sqrt{\frac{A_p^2}{\sigma_n^2}}\right)$
- $P_e = Q(\sqrt{SNR}) = Q(\sqrt{\gamma})$
- γ : SNR
- Minimizing the BER is same as maximizing the SNR.

Receiver: $f(t)$ (Minimum BER Criterion)

- $\gamma = \frac{p_0^2(t_m)}{\sigma_n^2}$
- $p_0(t)$?
- $p_0(t) = \int_{-\infty}^{\infty} P(f)F(f)e^{j2\pi ft}df$
- $p_0(t_m) = \int_{-\infty}^{\infty} P(f)F(f)e^{j2\pi ft_m}df$
- $\sigma_n^2 = \int_{-\infty}^{\infty} S_n(f)|F(f)|^2df$
- $\gamma = \frac{|\int_{-\infty}^{\infty} P(f)F(f)e^{j2\pi ft_m}df|^2}{\int_{-\infty}^{\infty} S_n(f)|F(f)|^2df}$

Receiver Structure: Matched Receiver

- Cauchy-Schwarz inequality:
$$|\int_{-\infty}^{\infty} \phi_1(x)\phi_2(x)dx|^2 \leq \int_{-\infty}^{\infty} |\phi_1(x)|^2 dx \int_{-\infty}^{\infty} |\phi_2(x)|^2 dx$$
- Equality $\phi_1(x) = k\phi_2^*(x)$
- $\phi_1(x) = F(f)\sqrt{S_n(f)}$, $\phi_2(x) = \frac{P(f)e^{j2\pi f t_m}}{\sqrt{S_n(f)}}$
- Apply Cauchy-Schwarz inequality:
- $\gamma \leq \int_{-\infty}^{\infty} \frac{|P(f)|^2}{S_n(f)} df$
- Maximized if $F(f)\sqrt{S_n(f)} = k[\frac{P(f)e^{j2\pi f t_m}}{\sqrt{S_n(f)}}]^*$
- $F(f) = k\frac{P(-f)e^{-j2\pi f t_m}}{S_n(f)}$
- Matched to pulse !

Matched Filtering: Condition

- Filtering = Receiver Structure
- $F(f) = k \frac{P(-f)e^{-j2\pi f t_m}}{S_n(f)}$
- Depends on: pulse shape, sampling instant, noise PSD

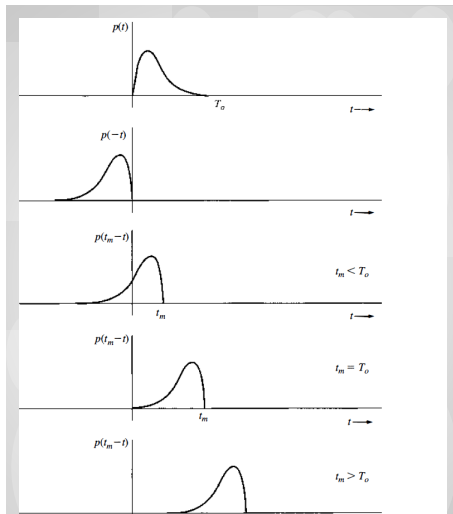
Matched Filtering: Gaussian Noise

- $F(f) = k \frac{P(-f)e^{-j2\pi ft_m}}{S_n(f)}$
- Consider Gaussian noise $S_n(f) = N_0/2$
- $F(f) = \frac{2k}{N_0} P(-f)e^{-j2\pi ft_m}$
- $f(t) = \frac{2k}{N_0} p(t_m - t)$
- Matched to pulse
- $\gamma_{max} = \int_{-\infty}^{\infty} \frac{|P(f)|}{S_n(f)} df$
- $\gamma_{max} = \frac{2}{N_0} \int_{-\infty}^{\infty} |P(f)|^2 df$
- $\gamma_{max} = \frac{2E_P}{N_0}$
- $P_e = Q(\sqrt{\frac{2E_P}{N_0}})$
- Pulse energy: $E_p = \int_{-\infty}^{\infty} |P(f)|^2 df = \int_{-\infty}^{\infty} |p(t)|^2 dt$
- BER is independent of pulse shape !

Matched Filtering: Sampling Instant

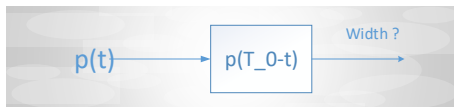
Pulse width = T_0 .

Which t_m is good?



Matched Filtering: Example

- If the width of $p(t)$ is T_0 , what is the width of MF $f(t)$?
- T_0
- What is the width at the output of MF?



- $2T_0$.

Matched Filtering: Example

Find the matched filter and output signal at the MF if the input signal pulse is a rectangular pulse of width T_0 .

General Binary Signaling

- Previously, we restricted to polar line code.
- General situation
- $y(t) = \begin{cases} p(t) + n(t), 0 \leq t \leq T_b & \text{bit 1} \\ q(t) + n(t), 0 \leq t \leq T_b & \text{bit 0} \end{cases}$
- $r(T_b) = q(T_b) + n(T_b)$ for bit $m = 0$
- $r(T_b) = p(T_b) + n(T_b)$ for bit $m = 1$
- $m = \begin{cases} 0, r < a_0 \\ 1, r > a_0 \end{cases}$

General Binary Signaling: BER

- $P(\epsilon|0) = P(r > a_0) = Q\left(\frac{a_0 - q_0(T_b)}{\sigma_n}\right)$
- $P(\epsilon|1) = P(r < a_0) = Q\left(\frac{p_0(T_b) - a_0}{\sigma_n}\right)$
- Average BER $P_e = \frac{1}{2}Q\left(\frac{a_0 - q_0(T_b)}{\sigma_n}\right) + \frac{1}{2}Q\left(\frac{p_0(T_b) - a_0}{\sigma_n}\right)$
- Optimum threshold: derivative of P_e with respect to a to zero
- Show that optimal threshold: $a_0 = \frac{p_0(T_b) + q_0(T_b)}{2}$
- Or, $a_0 = \frac{E_p - E_q}{2}$
- $f(t) = p(T_b - t) - q(T_b - t)$.
- $E_{pq} = \int_0^{T_b} p(t)q(t)dt$
- BER $P_e = Q\left(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}}\right)$

Receiver Structures

